



Presenter 1- Mumtahina Mehjabin Slides 01–05

Slide 01 - Title Slide

“Good morning everyone. Respected teacher and my dear classmates, Assalamu Alaikum.

Today, our group will present a research paper titled ‘**AmazonAICloud: Proactive Resource Allocation Using Amazon Chronos Based Time Series Model for Sustainable Cloud Computing.**’

This research focuses on **predicting future cloud resource demand and allocating resources more efficiently.**

I am **Mumtahina Mehjabin**, and I will begin with the abstract, introduction, and motivation.”

Slide 02 - Overview / Table of Contents

Here is the overall structure of our presentation.

Slide 03 - Abstract

“Now, coming to the abstract.

Cloud resource demand changes continuously, so reactive scaling may respond **after workload changes occur.**

To solve this problem, the framework uses **Amazon Chronos**, a pretrained time-series forecasting model, to predict future CPU utilization.

The study uses the **Bitbrains GWA-T-12 dataset**, containing traces from **1,750 virtual machines**, and forecasts **30 future time steps.**

Chronos is compared with **ARIMA, LSTM, and DeepAR** using **MAE, MSE, and MAPE**, as well as computational time.

The key finding is that **Chronos achieved the best reported forecasting error values.**”

Slide 04 - Introduction

“Now let's look at the introduction.

Cloud workloads are dynamic, meaning **CPU and memory demands continuously change.**

If too many resources are allocated, it causes **over-provisioning**, leading to wasted resources, cost, and energy.

If too few resources are allocated, it causes **under-provisioning**, which can reduce application performance and affect QoS or SLA requirements.

Therefore, cloud systems need to balance **performance, resource utilization, and efficiency.**”

Slide 05 - Motivation

“This brings us to the main motivation.

Traditional cloud management is usually **reactive**: it observes the workload, detects a threshold, and then scales resources.

The problem is that scaling may happen **after a workload spike has already occurred**.

Proactive management takes a different approach:

observe the past, forecast the future, and prepare resources in advance.

So, the main research question is:

Can accurate forecasting improve cloud resource-allocation decisions?

To investigate this, the researchers use **Amazon Chronos for future CPU prediction**.

Now, I would like to hand over the presentation to our team leader, **Md. Dulal Hossain**.



Presenter 2 - Md. Dulal Hossain Slides 06–12

Slide 06 - Research Contributions

“Thank you, Mumtahina.

Now I will discuss the main research contributions.

First, the study introduces **forecast-driven resource allocation** for proactive provisioning.

Second, it uses **Amazon Chronos**, a pretrained time-series model.

Third, the workload is analyzed using the **KPSS test**, especially for non-stationarity.

Fourth, Chronos is compared with **ARIMA, LSTM, and DeepAR**.

Finally, the experiment uses **AWS SageMaker** and connects forecasting with more efficient and sustainable cloud resource management.”

Slide 07 - Research Objectives

“The study has several important objectives.

First, to forecast future CPU utilization using historical VM workload data.

Second, to examine whether the workload is non-stationary using the **KPSS test**.

Third, to evaluate Amazon Chronos for cloud resource-demand forecasting.

Fourth, to compare Chronos with **ARIMA, LSTM, and DeepAR**.

Fifth, to measure forecasting performance using **MAE, MSE, and MAPE**.

And finally, to compare the computational efficiency of the models through their **training and inference times**.”

Slide 08 - Research Gap & Findings

“Now, let's look at the research gap.

Reactive scaling responds after workload changes, while traditional models may struggle with dynamic and non-stationary workloads.

Complex deep-learning models can also require significant training time.

To address these gaps, the study uses **pretrained Amazon Chronos**, real VM workload traces, and comparisons with established models. Its forecasts can then support **proactive resource provisioning**.”

■ Slide 09 - Methodology Overview

“The methodology follows six main steps:

First, collect VM workload data. **Second**, preprocess the data. **Third**, perform the **KPSS test**. **Fourth**, configure the forecasting models. **Fifth**, predict future CPU utilization. And **sixth**, evaluate accuracy and computational performance.

This provides evidence for using forecasting in proactive resource allocation.”

■ Slide 10 and 11 - System Architecture

“This slide shows the system architecture.

Historical **VM CPU data** is first processed and then provided to **Amazon Chronos**.

Chronos predicts future workload demand, and the forecast is passed to the **resource manager**.

The manager can then **provision or de-provision resources** based on predicted demand.

A feedback loop continuously adds actual utilization as new historical data.”

■ Slide 12 - Dataset Description

“The study uses the **Bitbrains GWA-T-12 cloud workload dataset**.

It contains traces from **1,750 virtual machines**, collected from **July 1 to October 1, 2013**.

The main variables include **CPU usage, CPU provisioned, CPU cores, memory usage, and memory provisioned**.

These data help analyze changing workload demand and resource utilization.”

■ Slide 13 - Amazon Chronos Algorithm

“Finally, let's look at the Chronos algorithm.

Historical CPU data is first **scaled and tokenized** into discrete representations.

Chronos then performs **autoregressive forecasting** and generates multiple forecast samples.

These samples are converted back into numerical CPU values.

In this experiment, the researchers use **Chronos Tiny**, with **30 forecast samples and a 30-step prediction length**.

With that, I hand over the presentation to **Bipro Roy**, who will discuss the equations, experimental setup, and results.”



Presenter 3 - Biproy Roy Slides 13–17

Slide 14 - Equations & Evaluation Metrics

“Thank you, Dulal.

This slide presents the three evaluation metrics: **MAE**, **MSE**, and **MAPE**.

MAE measures the average prediction error, **MSE** gives more importance to larger errors, and **MAPE** measures error in percentage.

For all three metrics, **lower values mean better performance.**”

Slide 15 - Experimental Setup

“The experiments were conducted using **AWS SageMaker Jupyter Notebook** with an **ml.t3.large CPU instance**.

The models predict the **next 30 time steps** using historical CPU utilization.

Four models were compared: **ARIMA**, **LSTM**, **DeepAR**, and **Amazon Chronos Tiny**.

Both **forecasting accuracy** and **computational efficiency** were evaluated.”

Slide 16 - Results: Forecast Accuracy

“Now let's look at the forecasting results.

ARIMA achieved MAE **13.825**, LSTM **8.217**, DeepAR **15.852**, while Chronos achieved the lowest MAE of **6.332**.

For MSE, Chronos achieved **91.390**, and for MAPE, **6.450**, which were also the lowest.

Therefore, **Chronos achieved the best overall forecasting accuracy** in this experiment.”

Slide 17 - Computational Efficiency

“Accuracy is important, but computational efficiency also matters.

ARIMA was the fastest, with only **0.03 seconds** training time.

LSTM took **40.8 seconds**, **DeepAR** **562 seconds**, and **Chronos** **324 seconds**.

Although ARIMA was faster, **Chronos provided significantly better forecasting accuracy**.

Compared with DeepAR, Chronos also required less training time. Now I will hand it over to **Sheikh Mehedi Hasan** .”



Presenter 4 - Sheikh Mehedi Hasan Slides 18–24



Slide 18 - Result Analysis

“Thank you, Bippro.

The results show that **Chronos achieved the lowest MAE, MSE, and MAPE**, making it the strongest overall model.

ARIMA was the fastest, while LSTM and DeepAR showed higher errors or computational costs.

Overall, **Chronos provides the best balance between accuracy and computational practicality.**”



Slide 19 - Discussion

“The results have several practical implications.

Accurate forecasting can give cloud managers **early information about future resource demand**.

This can support **proactive provisioning** and potentially reduce unnecessary resource allocation.

Since Chronos is pretrained, it can also simplify the adoption of forecasting models.

However, accurate forecasting alone is not enough; effective **scaling and resource-management policies** are still required.”



Slide 20 - Limitations & Conclusion

“There are some limitations.

The study mainly evaluates **forecasting performance**, while direct energy and carbon reductions were not measured.

Also, better forecasting does not automatically guarantee better resource allocation.

Therefore, real-world deployment requires suitable scaling policies and validation.

The main message is simple:

‘Predict first, allocate smarter.’”

Slide 21 - Future Work

“For future work, researchers can explore **hybrid ML models**, real-time cloud workload integration, and larger Chronos models.

They can also use **Generative AI for workload simulation** and directly optimize **energy and carbon emissions**.

Finally, **Explainable AI and SLA-aware resource management** can improve practical deployment.”

Slide 22 - References

“This slide presents the main reference used in our presentation.

The paper, ‘**AmazonAICloud: Proactive Resource Allocation Using Amazon Chronos Based Time Series Model for Sustainable Cloud Computing**,’ was published in *Computing*, Volume 107, Article 77, in **2025**.”

Slide 23 - Thank You / Q&A

“This brings us to the end of our presentation.

In summary, **Amazon Chronos can provide accurate workload forecasting to support proactive cloud resource management**.

Thank you very much for your attention.

We are now ready for your questions.”

Slide 24 - Final Closing

“Finally, thank you everyone for listening.

We hope our presentation provided a clear understanding of **Amazon Chronos, workload forecasting, proactive resource allocation, and sustainable cloud computing**.

We are happy to take your questions.

Thank you.”